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Solving Linear Equations

One method for every linear equation: clean it first, then isolate the letter. Brackets, fractions, letters on both sides — same two stages.

THE TOPIC ON ONE MAP

Solving Linear Equations

01 — Brackets and x on both sides

$$3(x - 2) = x + 7$$

$$5(4 - x) = 7 - 3x$$

$$4x - 13 = 17 + 8x$$

spot the mistake

02 — One fraction

$$\frac{5x - 3}{4} = 2x + 3$$

$$p = \frac{3p - 5}{10}$$

$$\frac{2f}{3} = 4f - 17$$

$$\frac{4(8x - 2)}{3x} = 10$$

03 — Two or more fractions

$$\frac{4x + 5}{3} - \frac{3 - 2x}{2} = 13$$

$$\frac{x + 2}{3x} + \frac{x - 2}{2x} = 3$$

$$3(x - 4) + \frac{x + 2}{5} = 6$$

04 — Hidden linear equations

$$(2x + 5)^2 = (2x + 3)(2x - 1)$$

$$fg(x) = f(x)$$

the x^2 cancels

IN THE EXAM Almost every Paper 1 and Paper 2 has one: part (b) or (c), 3 marks, “Show clear algebraic working”.

01 Clean, then Isolate

Every linear equation — however it looks — is solved in two stages. **Clean** removes the obstacles; **Isolate** gets x alone.

WHEN YOU SEE → YOU DO

“Solve” + one letter, no x^2

→ linear: Clean → Isolate

a bracket

→ expand it first (multiply **every** term inside)

x on both sides

→ move the **smaller** x term

“Show clear algebraic working”

→ write every line — trial and error scores 0

Clean

Collect

Add / Subtract

Divide

ZONE A

Brackets

Expand.

$$5(x + 3) \rightarrow 5x + 15$$

ZONE B

One fraction

Multiply both sides by the bottom.

$$\frac{5x - 3}{4} = \dots \rightarrow \times 4$$

ZONE C

Two fractions

Multiply **every term** by the LCM.

bottoms 3 and 2 → $\times 6$

1

Clean

Expand brackets, remove every denominator. Now there are no brackets, no fractions.

2

Collect

x terms on one side, numbers on the other. Change the sign when a term crosses “=”.

3

Add / Subtract, then Divide

Combine each side to one term, then divide by the number in front of x .

WORKED MODEL · BRACKETS AND x ON BOTH SIDESSolve $5(x + 3) = 3x - 4$.

1 $5x + 15 = 3x - 4$

clean: expand

2 $5x - 3x = -4 - 15$

collect (move $3x$)

3 $2x = -19$

add / subtract

ANSWER $x = -9.5$

× **TRAP** $5(x + 2) = 5x + 2$ is wrong — the 5 multiplies **both** terms: $5x + 10$.

IDEA 01

Practise: Clean, then Isolate

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Expand first

CH 2 · TOPIC 9 3 marks

Solve $4(x - 3) = 2x + 6$

X = _____

Q2 DR ESLAM Minus before a bracket

CH 2 · TOPIC 9 3 marks

Solve $7 - 2(x + 1) = 3x$

X = _____

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

CH 2 · TOPIC 9 5 marks

1 (a) Solve $3(x - 2) = x + 7$

(3 marks)

(b) Solve $\frac{2 - y}{5} = 1$

(2 marks)

Q4**PRACTICE**

Practice · Easy

CH 2 · TOPIC 9 3 marks

5 Solve $5x - 6 = 3(x - 1)$

(3 marks)

Q5**PRACTICE**

Practice · Easy

CH 2 · TOPIC 9 2 marks

10 Solve $5(x + 3) = 3x - 4$ Show clear algebraic working.

 $x =$

(2 marks)

Q6**PRACTICE**

Practice · Hard

CH 2 · TOPIC 9 7 marks

4 (a) Solve $5(f - 3) = f + 10$

(3 marks)

(b) Solve $\frac{h + 7}{3} + \frac{2h - 1}{2} = \frac{5}{6}$

(4 marks)

PAST PAPERS

Q7**PAST PAPER**

Edexcel 4MA1 · Oct/Nov 2020 · Paper 1HR · Q3

CH 2 · TOPIC 9**6 marks**

- 3 (a) Simplify $h^7 \times h^2$

.....

$$G = c^2 - 4c$$

- (b) Find the value of G when $c = -5$

$$G = \dots\dots\dots$$

- (c) Solve $\frac{5x - 3}{4} = 2x + 3$

Show clear algebraic working.

$$x = \dots\dots\dots$$

(Total for Question 3 is 6 marks)

Q8**PAST PAPER**

Edexcel 4MA1 · November 2024 · Paper 1H · Q3

CH 2 · TOPIC 9 6 marks

3 (a) Simplify $(p^3)^5$

.....

(b) Expand and simplify $2n(4n + 3) + n(n - 4)$

.....

(2)

(c) Solve $\frac{2x + 5}{3} = 4 - x$

Show clear algebraic working.

$x =$

(Total for Question 3 is 6 marks)

02 One Fraction: Multiply Both Sides

One denominator? Multiply **both sides** by it. Put the other side in a bracket first — that is where marks are lost.

WHEN YOU SEE → YOU DO

$$\frac{\square}{4} = \square$$

→ ×4 both sides, other side in a bracket

$$x \text{ in the denominator: } \frac{\square}{3x} = 10$$

→ still multiply both sides: ×3x

$$\text{"Solve } p = \frac{3p - 5}{10}\text{"}$$

→ $10p = 3p - 5$

Bracket the other side

Multiply up

Expand

Collect

Divide

WORKED MODEL • MAY 2024 P1H

$$\text{Solve } 3n - 4 = \frac{5n + 6}{3}.$$

$$1 \quad 3(3n - 4) = 5n + 6$$

clean: ×3, bracket

$$2 \quad 9n - 12 = 5n + 6$$

clean: expand

$$3 \quad 9n - 5n = 6 + 12$$

collect

$$4 \quad 4n = 18$$

add / subtract

ANSWER $n = 4.5$

× **TRAP** $3n - 4 = \frac{5n + 6}{3} \rightarrow 3n - 12 = 5n + 6$ is wrong: **every** term on that side is multiplied — $9n - 12$.

CALCULATOR CHECK Put your answer back in both sides (CALC button). Both sides equal? Done.

IDEA 02

Practise: One Fraction: Multiply Both Sides

My questions first, then selected practice, then real past papers.

Q9 DR ESLAM One fraction

CH 2 · TOPIC 9 3 marks

Solve $\frac{3x + 1}{4} = x - 2$

X = _____

Q10 DR ESLAM Fraction with a minus

CH 2 · TOPIC 9 3 marks

Solve $5 - \frac{x - 1}{3} = 2x$

X = _____

SELECTED PRACTICE

Q11 **PRACTICE** Practice · Easy

CH 2 · TOPIC 9 3 marks

8 Solve $y = \frac{2y + 1}{5}$

Show clear algebraic working.

$y =$
(3 marks)

Q12 **PRACTICE** Practice · Medium

CH 2 · TOPIC 9 3 marks

2 Solve $\frac{4(8x - 2)}{3x} = 10$

(3 marks)

Q13**PRACTICE**

Practice · Medium

CH 2 · TOPIC 9 3 marks

6 Solve $\frac{5 - x}{2} = 2x - 7$

(3 marks)

Q14**PRACTICE**

Practice · Medium

CH 2 · TOPIC 9 3 marks

13 Solve $\frac{2x + 5}{6} = 2x - 5$

Show clear algebraic working.

(3 marks)

PAST PAPERS

Q15**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1H • Q7

CH 2 • TOPIC 9 9 marks7 (a) Expand and simplify $(m - 8)(m + 5)$

.....

(b) Factorise fully $5y + 20y^2$

.....

(c) Simplify $(p^2 + 3)^0$

.....

(1)

(d) Solve $3(2x - 5) = \frac{9 - x}{2}$

Show clear algebraic working.

 $x =$ **(Total for Question 7 is 9 marks)**

Q16**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 1H · Q5

CH 2 · TOPIC 9 5 marks

5 (a) Factorise fully $15y^4 + 20uy^3$

.....
(2)

(b) Solve $4 - 3x = \frac{5 - 8x}{4}$

Show clear algebraic working.

$x =$

(Total for Question 5 is 5 marks)

Q17**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 2HR · Q3

CH 2 · TOPIC 9 5 marks3 (a) Solve $4y + 5 > 12$
(2)(b) Solve $6x - 5 = \frac{4x - 7}{2}$

Show clear algebraic working.

 $x =$ **(Total for Question 3 is 5 marks)**

Q18 **PAST PAPER** Edexcel 4MA1 · June 2022 · Paper 1H · Q5**CH 2 · TOPIC 9** **5 marks**5 (a) Expand and simplify $(n - 6)(n + 4)$

(b) Solve $2x - 3 = \frac{3x - 5}{4}$

Show clear algebraic working.

$x = \dots\dots\dots$

(Total for Question 5 is 5 marks)

Q19**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 2HR · Q6

CH 2 · TOPIC 9 8 marks

6 (a) Solve $p = \frac{3p - 5}{10}$

Show clear algebraic working.

$$p = \dots\dots\dots$$

(b) Simplify a^0 where $a > 0$

$$\dots\dots\dots (1)$$

(c) Simplify fully $\frac{3xy^3}{6x^2y}$

$$\dots\dots\dots$$

(d) Factorise fully $10c^3d^2 + 15cd^4$

$$\dots\dots\dots$$

(Total for Question 6 is 8 marks)

Q20**PAST PAPER**

Edexcel 4MA1 · June 2024 · Paper 2HR · Q6

CH 2 · TOPIC 9 8 marks

6 (a) Solve $\frac{2f}{3} = 4f - 17$

Show clear algebraic working.

$$f = \dots\dots\dots$$

(b) Simplify $(e + 12)^0$ where $e > 0$

$$\dots\dots\dots (1)$$

(c) Simplify fully $\frac{12a^4h^6}{4ah^2}$

$$\dots\dots\dots$$

(d) Factorise fully $20x^5y + 12x^3y^4$

$$\dots\dots\dots$$

(Total for Question 6 is 8 marks)

Q21**PAST PAPER**

Edexcel 4MA1 · May 2024 · Paper 1H · Q6

CH 2 · TOPIC 9 5 marks6 (a) Expand and simplify $(m + 5)(m - 8)$ (b) Solve $3n - 4 = \frac{5n + 6}{3}$

Show clear algebraic working.

 $n = \dots\dots\dots$ **(Total for Question 6 is 5 marks)**

Q22**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2HR · Q1

CH 2 · TOPIC 9 5 marks

- 1 (a) Factorise fully $18c - 45cd$

.....
(2)

- (b) Solve $\frac{5 - 2x}{6} = 3x - 4$

Show clear algebraic working.

$x =$

(Total for Question 1 is 5 marks)

03 Two or More Fractions: LCM

Several denominators? Find the LCM and multiply **every** term — including the number on the right.

WHEN YOU SEE → YOU DO

$$\frac{\square}{3} - \frac{\square}{2} = 13$$

→ LCM = 6: $2(\dots) - 3(\dots) = 78$

a minus in front of a fraction

→ bracket it: $-3(3 - 2x) = -9 + 6x$

x in the bottoms: 3x, 2x

→ LCM = 6x

a term with no fraction

→ it gets multiplied too

Find LCM

Multiply every term

Cancel

Expand

Collect

Divide

WORKED MODEL • JUNE 2022 P2H

Solve $\frac{4x+5}{3} - \frac{3-2x}{2} = 13$.

1 $2(4x+5) - 3(3-2x) = 78$

clean: $\times 6$

2 $8x + 10 - 9 + 6x = 78$

expand (watch the -)

3 $14x + 1 = 78 \Rightarrow 14x = 77$

collect

ANSWER $x = 5.5$

× **TRAP** Forgetting to multiply the right-hand side: $2(4x+5) - 3(3-2x) = 13$ loses the method mark. It is $13 \times 6 = 78$.

IDEA 03

Practise: Two or More Fractions: LCM

My questions first, then selected practice, then real past papers.

Q23

DR ESLAM

LCM of 2 and 5

CH 2 · TOPIC 9 4 marks

Solve $\frac{x+3}{2} - \frac{x-1}{5} = 4$

X = _____

Q24

DR ESLAM

Every term

CH 2 · TOPIC 9 3 marks

Solve $\frac{x}{3} + \frac{x}{4} = 14$

X = _____

SELECTED PRACTICE

Q25

PRACTICE

Practice · Hard

CH 2 · TOPIC 9 3 marks

1 Solve $\frac{4x - 1}{5} + \frac{x + 4}{2} = 3$

(3 marks)

Q26

PRACTICE

Practice · Hard

CH 2 · TOPIC 9 3 marks

3 Solve $\frac{x + 2}{3x} + \frac{x - 2}{2x} = 3$

(3 marks)

Q27**PRACTICE**

Practice · Hard

CH 2 · TOPIC 9 3 marks

6 Solve $\frac{9a-7}{5} - \frac{3a-7}{4} = 4.55$

Show clear algebraic working.

a =

(3 marks)

Q28**PRACTICE**

Practice · Hard

CH 2 · TOPIC 9 4 marks

10 Solve the equation.

$$3(x-4) + \frac{x+2}{5} = 6$$

x =

(4 marks)

PAST PAPERS

Q29**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2H • Q15

CH 2 • TOPIC 9 7 marks

15 (a) Solve $\frac{4x+5}{3} - \frac{3-2x}{2} = 13$

Show clear algebraic working.

$x =$

(b) Solve the inequality $2y^2 - 7y - 30 \leq 0$
Show your working clearly.

.....

(Total for Question 15 is 7 marks)

Q30**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 1HR · Q15

CH 2 · TOPIC 9 6 marks

15 (a) Solve $\frac{5a+8}{3} - \frac{2a+5}{4} = 23$

Show clear algebraic working.

$$a = \dots\dots\dots$$

(4)

(b) Express $\left(\frac{\sqrt{y}}{3}\right)^{-1}$ in the form cy^n where c and n are numbers to be found.

(Total for Question 15 is 6 marks)

04 Hidden Linear Equations

Some equations *look* quadratic. Expand everything — if the x^2 terms cancel, it was linear all along.

WHEN YOU SEE → YOU DO

$$(2x + 5)^2 = (2x + 3)(2x - 1)$$

→ expand both sides — the $4x^2$ cancels

$$fg(x) = f(x)$$

→ build $fg(x)$, then the x^2 cancels

“What mistake did he make?”

→ find the first wrong line, say what should be there

WORKED MODEL • THE x^2 CANCELS

Solve $(2x + 5)^2 = (2x + 3)(2x - 1)$.

$$1 \quad 4x^2 + 20x + 25 = 4x^2 + 4x - 3$$

clean: expand both

$$2 \quad 20x - 4x = -3 - 25$$

$4x^2$ cancels; collect

$$3 \quad 16x = -28$$

add / subtract

ANSWER $x = -\frac{7}{4}$

WHERE YOU WILL USE IT Rearranging formulas (Topic 7), simultaneous equations (Topic 12), forming equations (Topic 13) all end in a linear equation.

IDEA 04

Practise: Hidden Linear Equations

My questions first, then selected practice.

Q31 DR ESAM The square cancels

CH 2 · TOPIC 9 3 marks

Solve $(x + 4)^2 = x^2 + 40$

$x =$ _____

SELECTED PRACTICE**Q32** PRACTICE Practice · Easy

CH 2 · TOPIC 9 1 mark

6 Steve is asked to solve the equation $5(x + 2) = 47$

Here is his working.

$$5(x + 2) = 47$$

$$5x + 2 = 47$$

$$5x = 45$$

$$x = 9$$

Steve's answer is wrong.

What mistake did he make?

(1 mark)

Q33**PRACTICE**

Practice · Hard

CH 2 · TOPIC 9 3 marks

7 Solve $(2x + 5)^2 = (2x + 3)(2x - 1)$

 $x = \dots\dots\dots$ **(3 marks)****Q34****PRACTICE**

Practice · Hard

CH 2 · TOPIC 9 3 marks

9 The functions f and g are defined as

$$f(x) = x^2 + 6$$

$$g(x) = x - 10$$

Solve the equation $fg(x) = f(x)$

Show clear algebraic working.

(3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.



- 01 Clean, then Isolate** Expand brackets → collect (move the smaller x) → add/subtract → divide.
- 02 One fraction** Bracket the other side, multiply both sides by the bottom.
- 03 Two fractions** LCM × **every** term, the right-hand side too. Bracket after a minus.
- 04 Hidden linear** Expand everything; the x^2 cancels.

BOTTOM LINE Clean → Collect → Add / Subtract → Divide → check with CALC.

Answers

Solving Linear Equations · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · CLEAN, THEN ISOLATE

- | | | |
|-----------|--|--|
| Q1 | Dr Eslam · Expand first | $4x - 12 = 2x + 6; x = 9$ |
| Q2 | Dr Eslam · Minus before a bracket | $7 - 2x - 2 = 3x; x = 1$ |
| Q3 | Practice · Easy | (a) $x = 6.5$ (b) $y = -3$ |
| Q4 | Practice · Easy | $x = 1.5$ |
| Q5 | Practice · Easy | $x = -9.5$ |
| Q6 | Practice · Hard | (a) $f = \frac{25}{4}$ (b) $h = -\frac{3}{4}$ |
| Q7 | Edexcel 4MA1 · Oct/Nov 2020 · Paper 1HR · Q3 | (a) h^9 (b) 45 (c) $x = -5$ |
| Q8 | Edexcel 4MA1 · November 2024 · Paper 1H · Q3 | (a) p^{15} (b) $9n^2 + 2n$ (c) $x = \frac{7}{5}$ |

IDEA 02 · ONE FRACTION: MULTIPLY BOTH SIDES

- Q9** Dr Eslam · One fraction $3x + 1 = 4x - 8; x = 9$
- Q10** Dr Eslam · Fraction with a minus $15 - (x - 1) = 6x; x = \frac{16}{7}$
- Q11** Practice · Easy $y = \frac{1}{3}$
- Q12** Practice · Medium $x = 4$
- Q13** Practice · Medium $x = 3.8$
- Q14** Practice · Medium $x = 3.5$
- Q15** Edexcel 4MA1 · January 2020 · Paper 1H · Q7 (a) $m^2 - 3m - 40$ (b) $5y(1 + 4y)$ (c) 1 (d) $x = 3$
- Q16** Edexcel 4MA1 · January 2021 · Paper 1H · Q5 (a) $5y^3(3y + 4u)$ (b) $x = \frac{11}{4}$
- Q17** Edexcel 4MA1 · January 2022 · Paper 2HR · Q3 (a) $y > \frac{7}{4}$ (b) $x = \frac{3}{8}$
- Q18** Edexcel 4MA1 · June 2022 · Paper 1H · Q5 (a) $n^2 - 2n - 24$ (b) $x = \frac{7}{5}$
- Q19** Edexcel 4MA1 · June 2022 · Paper 2HR · Q6 (a) $p = -\frac{5}{7}$ (b) 1 (c) $\frac{y^2}{2x}$ (d) $5cd^2(2c^2 + 3d^2)$
- Q20** Edexcel 4MA1 · June 2024 · Paper 2HR · Q6 (a) $f = 5.1$ (b) 1 (c) $3a^3h^4$ (d) $4x^3y(5x^2 + 3y^3)$
- Q21** Edexcel 4MA1 · May 2024 · Paper 1H · Q6 (a) $m^2 - 3m - 40$ (b) $n = 4.5$
- Q22** Edexcel 4MA1 · June 2025 · Paper 2HR · Q1 (a) $9c(2 - 5d)$ (b) $x = 1.45$

IDEA 03 · TWO OR MORE FRACTIONS: LCM

- Q23** Dr Eslam · LCM of 2 and 5 $5(x + 3) - 2(x - 1) = 40; x = \frac{23}{3}$
- Q24** Dr Eslam · Every term $4x + 3x = 168; x = 24$
- Q25** Practice · Hard $x = \frac{12}{13}$
- Q26** Practice · Hard $x = -\frac{2}{13}$
- Q27** Practice · Hard $a = 4$
- Q28** Practice · Hard $x = 5.5$
- Q29** Edexcel 4MA1 · June 2022 · Paper 2H · Q15 (a) $x = 5.5$ (b) $-2.5 \leq y \leq 6$
- Q30** Edexcel 4MA1 · June 2025 · Paper 1HR · Q15 (a) $a = 18.5$ (b) $3y^{-\frac{1}{2}}$

IDEA 04 · HIDDEN LINEAR EQUATIONS

- Q31** Dr Eslam · The square cancels $x^2 + 8x + 16 = x^2 + 40; x = 3$
- Q32** Practice · Easy He wrote $5x + 2$: it should be $5x + 10$ (so $x = 7.4$)
- Q33** Practice · Hard $x = -\frac{7}{4}$
- Q34** Practice · Hard $(x - 10)^2 + 6 = x^2 + 6; x = 5$